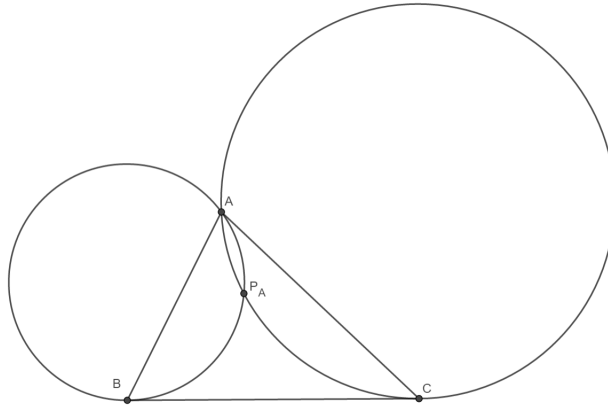


1 Humpty Point

The **Humpty point** is defined as follows:

Let Ω_1 be the circle passing through A and tangent to BC at B . Let Ω_2 be the circle passing through A and tangent to BC at C . Then the two circles meet at A and the A -Humpty point, denoted by P_A .



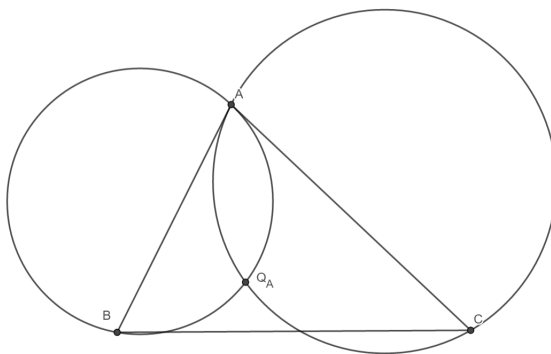
In other words, P_A is the point inside the triangle such that $\angle P_A BC = \angle P_A AB$ and $\angle P_A CB = \angle P_A AC$.

1. A , P_A and M are collinear, where M is the midpoint of segment BC .
2. (a) $\frac{AB}{AC} = \frac{P_A B}{P_A C}$.
 (b) (A -Appolonius circle) A , P_A , D and E are concyclic, where D and E are points on BC such that AD is the internal angle bisector of $\angle BAC$ and AE is the external angle bisector of $\angle BAC$.
3. B , P_A , H , C are concyclic, where H is the orthocentre of $\triangle ABC$.
4. (a) P_A lies on the circle with diameter AH .
 (b) $AP_A \cdot AM = bc \cos A$.
 (c) P_A is the image of M under the inversion with radius $\sqrt{AH \cdot AH_A}$ and centre A , where H_A is the perpendicular foot from A to BC .
5. P_A , M , O' , F are concyclic, where O' is the reflection of the circumcentre O of $\triangle ABC$ about line BC and F is the midpoint of AH .
6. P_A and S are of reflection symmetry about line BC , where S is the intersection of the A -symmedian (A -median reflected about A -angle bisector) and the circumcircle of $\triangle ABC$.

2 Dumpty Point

The **Dumpty point** is defined as follows:

Let Γ_1 be the circle passing through B and tangent to AC at A . Let Γ_2 be the circle passing through C and tangent to AB at A . Then the two circles meet at A and the A -Dumpty point, denoted by Q_A .



In other words, Q_A is the point inside the triangle such that $\angle Q_A B A = \angle Q_A A C$ and $\angle Q_A A B = \angle Q_A C A$.

1. Q_A lies on the A -symmedian.
2. The Humpty Dumpty points are isogonal conjugates in $\triangle ABC$.
3. Q_A is the centre of spiral similarity sending AC to BA .
4. B, Q_A, O, C are concyclic.
5. (a) Q_A lies on the circle with diameter AO .
(b) Q_A is the midpoint of AS , where S is the intersection of the A -symmedian and the circumcircle of $\triangle ABC$.

3 Exercises

1. In $\triangle ABC$, P is a variable point on the segment BC . Let M and N be points on AB and AC respectively such that $PM \parallel AC$ and $PN \parallel AB$. Prove that as P varies on BC , the circumcircle of $\triangle AMN$ passes through a fixed point.
2. In a semi-circle with centre O and diameter AB , let M and N be points on the semicircle, and let X be the intersection of AB and NM . Suppose circumcircles of $\triangle MBO$ and $\triangle NAO$ meet at K and O . Prove that $XK \perp KO$.
3. Let P be a point on the A -symmedian of $\triangle ABC$. Let O, O_1 and O_2 be the circumcentres of $\triangle ABC$, $\triangle APB$ and $\triangle CAP$ respectively. Prove that AO bisects O_1O_2 .
4. In $\triangle ABC$, let O be the circumcentre and D be the perpendicular foot from A to BC . A circle with centre on AD is tangent externally to the circumcircle of $\triangle BOC$ at X . Prove that AX is the A -symmedian.

4 Problems

- (IMO Shortlist 2006) Let $ABCDE$ be a convex pentagon such that $\angle BAC = \angle CAD = \angle DAE$ and $\angle ABC = \angle ACD = \angle ADE$. The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .
- (ELMO 2014) Let ABC be a triangle with circumcentre O and orthocentre H . Let ω_1 and ω_2 denote the circumcircles of $\triangle BOC$ and $\triangle BHC$ respectively. Suppose the circle with diameter AO intersects ω_1 again at M , and line AM intersects ω_1 again at X . Similarly, suppose the circle with diameter AH intersects ω_2 again at N , and line AN intersects ω_2 again at Y . Prove that lines MN and XY are parallel.
- (USAMO 2008) Let $\triangle ABC$ be an acute, scalene triangle, and let M, N and P be the midpoints of BC, CA and AB respectively. Let the perpendicular bisectors of AB and AC intersect line AM at points D and E respectively, and let lines BD and CE intersect at point F . Prove that A, N, F, P are concyclic.
- (EGMO 2016) Two circles ω_1 and ω_2 of equal radius intersect at different points X_1 and X_2 . Consider a circle ω externally tangent to ω_1 at T_1 and internally tangent to ω_2 at point T_2 . Prove that lines X_1T_1 and X_2T_2 intersect at a point lying on ω .
- (USA TSTST 2015) Let ABC be a scalene triangle. Let K_a, L_a and M_a be the respective intersections with BC of the A -internal angle bisector, A -external angle bisector and the A -median. The circumcircle of AK_aL_a intersects AM_a a second time at point X_a different from A . Define X_b and X_c analogously. Prove that the circumcenter of $X_aX_bX_c$ lies on the Euler line (the line passing through the circumcentre, centroid and orthocentre) of $\triangle ABC$.
- (ELMO Shortlist 2013) In $\triangle ABC$, a point D lies on line BC . The circumcircle of $\triangle ABD$ meets AC at F (other than A), and the circumcircle of $\triangle ADC$ meets AB at E (other than A). Prove that as D varies, the circumcircle of $\triangle AEF$ always passes through a fixed point other than A , and that this point lies on the median from A to BC .
- (USA TST 2005) Let ABC be an acute scalene triangle with O as its circumcentre. Point P lies inside $\triangle ABC$ with $\angle PAB = \angle PBC$ and $\angle PAC = \angle PCB$. Point Q lies on line BC with $QA = QP$. Prove that $\angle AQP = 2\angle OQB$.
- (IMO Shortlist 2019 modified) In $\triangle DEF$, let I be the circumcentre, and let A, B, C be the midpoints of EF, FD, DE respectively. Let the circumcircles of $\triangle ABC$ and $\triangle IEF$ meet at P and Q . Prove that $\angle PDQ + \angle PAQ = \angle PIQ$.

